

QoE-based Cross-layer Optimization for Uplink Video Transmission

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We study the problem of resource-efficient uplink distribution of user-generated video content over fourth generation mobile networks. This is challenged by 1) the capacity-limited and time-variant uplink channel, 2) the resource-hungry upstreamed videos and their dynamically changing complexity, and 3) the different playout times of the video consumers. To address these issues, we propose a systematic approach for quality of experience (QoE) based resource optimization and uplink transmission of multi-user generated video content. More specifically, we present an analytical model for distributed scalable video transmission at the mobile producers which considers the above constraints. This is complemented by a multi-user cross-layer optimizer in the mobile network which determines the transmission capacity for each mobile terminal under current cell load and radio conditions. Both optimal and low-complexity solutions are presented. Simulation results for LTE uplink transmission show that significant gains in perceived video quality can be achieved by our cross-layer resource optimization scheme. In addition, the distributed optimization at the mobile producers can further improve the user experience across the different types of video consumers.

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General Terms: Algorithms, Design

Additional Key Words and Phrases: Quality of Experience, live and VoD, LTE uplink, scalable video transmission, resource allocation

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1. INTRODUCTION

Mobile video is expected to account for more than two-thirds of mobile data traffic by 2018 [Cisco white paper 2014]. This development is mainly driven by a rapid increase in smartphone users. Smartphone users are both producers and consumers of media content thanks to the enhanced camera capabilities of mobile devices [Gabin et al. 2010]. As a result, upstreaming of rich multimedia content from mobile devices to video portals (e.g., YouTube) is increasingly common nowadays. Moreover, camcorders, in-

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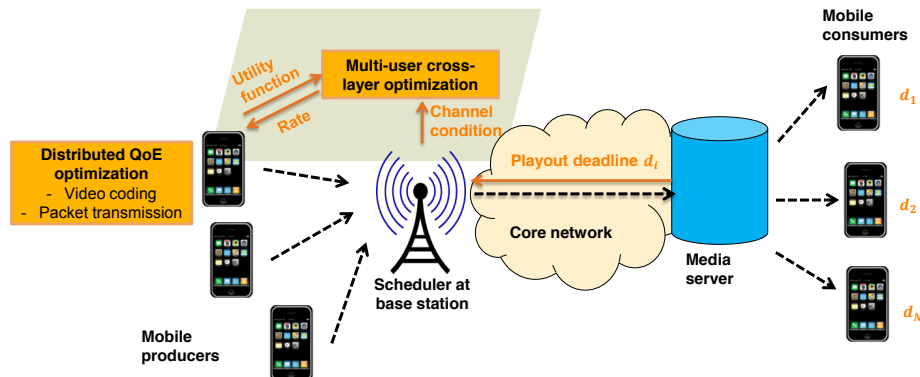


Fig. 1. System image for QoE-driven uplink resource optimization, which includes a multi-user cross-layer optimization in the mobile network and a distributed QoE optimization at the mobile terminals.

car cameras and mobile users on the street are generating large volumes of video data which are being uploaded over the mobile network. Mobile devices including smartphones and tablets will play a central role in caching and distributing multimedia content in next generation mobile networks [Boccardi et al. 2014].

Despite the capacity enhancements in next generation wireless networks, they can not meet the progression in the mobile media market [Oyman et al. 2010]. The down-link channel has been considered the main bottleneck in the past. Nevertheless, the increasing availability of smartphones will tighten the burden on the capacity-limited uplink distribution channel. More specifically, the ultimate goal of a mobile network operator is to maximize the users' quality of experience (QoE) [Akamai Technologies 2015]. The admission control mechanisms implemented in mobile networks are based on static Quality of Service (QoS) constraints that map application-specific information into QoS parameters (e.g., QoS Class Identifiers (QCIs), guaranteed bit-rate (GBR)) [3GPP TS 29.213 v12.4.0 2014] [3GPP TS 29.214 v12.3.0 2014]. These mapping functions are not suited for the dynamic and variable characteristics of video contents. Moreover, the majority of video streaming users use non-GBR bearers [3GPP TR 23.705 2013]. Furthermore, QoS-aware schedulers [Capozzi et al. 2013] that map the QoS constraints into QoS parameters such as minimum delay, guaranteed bit-rate or packet loss rate are content agnostic and thus do not truly reflect the user satisfaction. To this end, it has become increasingly important for mobile operators to look for user-centric approaches that complement the QoS mechanisms in the network.

In this paper, we address the following question: how to efficiently upload user generated video content from mobile terminals in next generation mobile networks? We consider a mobile cell where multiple videos need to be simultaneously uploaded from mobile terminals to a media server. We are particularly interested in live and near real-time users that consume the uploaded videos either immediately or within several seconds. In this context, it is important to address the playout requirements of the uploaded videos. For instance, not all videos will be watched at the same time or enjoy the same viewing popularity [El Essaili et al. 2011a]. Uploading a high quality video content might not be always feasible in real-time in a resource-constraint and variable uplink spectrum. Scalable video coding (SVC), on the other hand, encodes a video into a base layer (BL) and several enhancement layers (EL) [Schwarz et al. 2007]. Our approach is to progressively upload high quality video content by

utilizing the variability in the uplink communication channel. To this end, we exploit the local caching capabilities of mobile terminals to store parts of the video stream which cannot be instantaneously uploaded due to bandwidth limitations. A schematic overview of our proposed system for uplink resource optimization is illustrated in Figure 1. Multiple mobile video producers are upstreaming their content to a media server. In particular, the server collects feedback from consumers on their playout time (deadlines $d_i, i = 1 \dots N$) and provides this feedback to the video producers. The uplink resource allocation is then optimized to provide video consumers with the best possible QoE within the scheduled playout time taking the limited and time-varying uplink resources into account. Our approach includes a multi-user cross-layer optimizer (CLO) in the mobile network (close to the base station) which is responsible for the uplink resource allocation among the video producers such that the overall QoE is maximized. Complementing it, a distributed QoE-based optimization is performed by each video producer to decide on which video packets to transmit and their respective rates.

In summary, our contribution is twofold: On one hand, we provide a solution for the optimization of the uplink scalable video transmission over time at the mobile terminals while taking into account the dynamics of the mobile network and the consumers' characteristics. In this context, both optimal and low-complexity algorithms are proposed and evaluated. On the other hand, we propose a multi-user CLO to determine the transmission capacity for each mobile terminal such that the wireless resources are efficiently utilized. The cross-layer optimizer acts as an overlay on top of the standard scheduler in the mobile network which is responsible for the actual resource allocation among the video producers.

The rest of the paper is organized as follows. In the next section, we first review related work. Section 3 outlines our system model. In Section 4, we present our distributed QoE optimization which is performed by all terminals individually. Section 5 describes our CLO framework for uplink resource allocation among multiple video producers. Section 6 presents the simulation results and Section 7 concludes the paper.

2. RELATED WORK

2.1. Background

Optimizing the video delivery over wireless communication channels has been thoroughly studied in the literature. Mobile media streaming has evolved from simple RTP/UDP to adaptive HTTP/TCP to cope with the variability of the mobile networks [Gabin et al. 2010]. Despite these major advances in multimedia delivery protocols, they generally provide an end-to-end adaptation mechanism whereas multi-access and network awareness remain an open issue [Begen et al. 2011]. Multi-stream video (e.g., Multiple Description Coding [Goyal 2001] and Scalable Video Coding [Schwarz et al. 2007]) encodes the same content into multiple representations and provides a more flexible framework for in-network adaptation (e.g., description or layer dropping). Loss adaptive media coding that combines multi-stream video with forward error correction has been studied as well and results in higher resilience against network variations (e.g., [Hamzaoui et al. 2005]).

Cross-layer approaches have also been considered to optimize the video delivery by exchanging information across the different protocol layers [Berry and Yeh 2004]. Conventional CLO adapts the instantaneous transmission parameters on a short timescale which is not optimal from a multimedia quality perspective [van der Schaar and Sai Shankar 2005]. Application-driven cross-layer optimization, on the other hand, directly maximizes an application-specific objective function while using abstracted models for the link layer and physical layer [Khan et al. 2006]. Recently, QoE-driven CLO has been proposed for downlink resource allocation in HSDPA networks [Thakolsri

et al. 2009]. Long-term adaptation cycles (order of seconds) based on average channel characteristics are considered in [Thakolsri et al. 2009] to maximize the user QoE. Meanwhile, uplink resource management, addressed in this paper, brings along several challenges compared to the downlink case. First, for downlink-based services, a pure network-centric approach for optimization is possible as both the network including the base station and the content source are in front of the bottleneck wireless link. However, for uplink-based services the bottleneck is between the content source, i.e., the terminal and the network. Hence, a distributed optimization approach has to be considered involving entities in the network and in the terminal. Moreover, different from the downlink scenario where the resource allocation maximizes the QoE of the mobile users downloading the video content, the end goal of the uplink resource allocation is to maximize the perceived quality of the users watching the upstreamed videos. The challenge is to schedule the data packets at the mobile terminals which need to be uploaded over the wireless uplink channel. Consequently, some knowledge about the video consumers is required to optimize the uplink distribution. Our approach is to include the consumers' requirements into the uplink optimization problem, which is available through feedback from a media server or portal.

2.2. Scheduling with deadlines

Earliest Deadline First (EDF) [Georgiadis et al. 1997] is one of the first studied scheduling policies whereby the objective is to schedule the packets with the most imminent deadline. Packets delivered after their deadline are useless and therefore the scheduler aims at delivering each packet before its expiry time. [Shakkottai and Srikant 2002] shows that the EDF scheduling policy is sub-optimal in mobile environments as it fails to exploit the channel variabilities. The round robin, maximum throughput and proportional fair (PF) scheduling schemes have been considered to achieve different levels of fairness/throughput efficiency in a mobile network [Dimitrova et al. 2011]. The above scheduling mechanisms, however, are not optimized for handling real-time multimedia communications with strict delay constraints. On the one hand, the scheduler in the mobile network is typically unaware of the individual packet requirements. On the other hand, the scheduling decisions are performed on a short-term transmission time interval (TTI) basis which is not optimal from a multimedia perspective.

The seminal work on packetized media transmission, which takes into account the content characteristics and the decoding deadlines of the individual packets, goes back to [Chou and Miao 2006]. In their approach, the dependencies among the media packets are captured by a directed acyclic graph. This allows to prioritize the packet transmission based on the R-D characteristics and the relative importance of the packets. In [Chou and Miao 2006], the channel condition is modeled with a Markov Decision Process (MDP) and assumed to be time-invariant where the transmission policy for a group of packets can be determined apriori. This is different from the time-varying characteristics of wireless channels where the transmission policies should be determined at each scheduling interval. The original work from [Chou and Miao 2006] has been extended in several directions to account for channel, deadline and distortion impacts in multi-user streaming scenarios (e.g., [Chakareski and Frossard 2006]).

A cross-layer packetization and retransmission approach for video transmission in wireless LANs that explicitly considers the actual transmission requirements is proposed in [van der Schaar and Turaga 2007]. More specifically, the video packets are grouped by their decoding deadlines and the cross-layer optimization problem is solved independently for all packets with a common decoding deadline using a general Lagrangian optimization formulation. While the authors do not consider the potential benefit of transmitting some packets with a later deadline before those with an earlier

deadline, it is favorable in our case to schedule layers with a later deadline to guarantee a basic quality in real-time before previously cached enhancement layers with an earlier deadline are transmitted for on-demand consumption. As a result, the optimization problem cannot be solved independently as proposed in [van der Schaar and Turaga 2007]. In [Dua et al. 2010], a mathematical framework for channel, deadline and distortion aware scheduling is proposed. The work follows the principle of low-latency media system design where packets are associated with strict deadlines, but it does not further investigate the possibility of sending queued packets for time-shifted consumption.

Much research work has investigated scalable video transmission over error-prone fading channels. In [Khalek et al. 2012], the authors present a cross-layer framework for optimizing the perceptual video quality in scalable video transmission over a Rayleigh fading channel. A joint source-channel formulation is described to jointly adapt the layer scheduling and the modulation and coding scheme (MCS) subject to playback buffer underflow and packet error rate constraints. [Lin et al. 2012] deals with the problem of scalable video transmission over error-prone channels with a tight delay constraint. An optimization problem is formulated to minimize the distortion by looking for the appropriate payload length, video rate and MCS mode. Scalable video transmission over fading wireless channels is also considered in [Zhang et al. 2010] with the objective of maximizing the average video quality. Video frames are scheduled in the order of decoding deadlines and layers that exceed the rate constraint are dropped. [Cicalo and Tralli 2014] studies scalable video streaming over OFDMA channels with the objective of achieving fairness among the users. Different from the above previous works, we study scalable video transmission with the objective of optimizing the video quality for both live and on-demand video consumers.

Several research works have studied deadline-aware media transmission in LTE networks. The authors in [Ji et al. 2009] propose a multi-user gradient-based scheduling framework that considers the streaming of scalable videos with different playback deadlines in an OFDMA system. They present a dynamic weighting metric for mitigating the approaching deadline effect such that video layers with imminent deadlines are assigned a higher priority. Again, the idea is to enforce the scheduler to assign more resources for users with approaching deadlines without considering the possibility of storing and transmitting the video layers for on-demand retrieval. A two-level scheduling approach for real-time media delivery in LTE networks is proposed in [Piro et al. 2011]. More specifically, a long-term resource allocation algorithm runs on top of the PF scheduler to decide the amount of data that each source should transmit in order to satisfy its delay constraint. The main drawback of the approach in [Piro et al. 2011] is that the upper level scheduler does not take into account the channel conditions of the users. Therefore, there is no guarantee that the PF scheduler can assign each user the amount of resources it needs to satisfy its delay requirements. Different from [Piro et al. 2011], the approach presented in this paper leverages cross-layer information about the channel conditions and video characteristics of each user to determine its optimal rate budget.

3. SYSTEM MODEL

3.1. Application model

We consider a QoE-driven approach for uplink resource allocation. The user satisfaction is expressed on a Mean Opinion Score (MOS) scale [ITU 1996]. Specifically, we are interested in characterizing the utility for video streaming which is defined in [Khan et al. 2007] as a function of the application data rate R by:

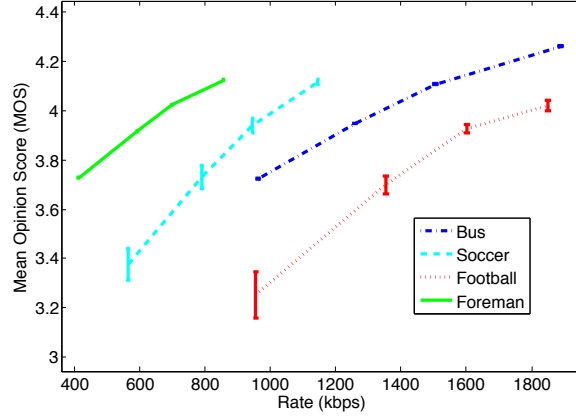


Fig. 2. Mean Opinion Score (MOS) as a function of data rate. The figure shows the average MOS and the variance over 20 GoPs. The test video sequences are encoded with the H.264 SVC codec with constant quality, at CIF resolution, a GoP size of 8 frames, and a rate of 30 frames/sec.

$$U = f(R), f : R \rightarrow MOS \quad (1)$$

A linear mapping between the Structural SIMilarity (SSIM) and MOS is considered, which is based on the scatter plot of subjective and objective results for the SSIM video quality metric [Wang et al. 2004]:

$$MOS = \begin{cases} 1 & \text{if } SSIM < 0.76 \\ 15.91 \cdot SSIM - 11.09 & \text{if } 0.76 \leq SSIM \leq 0.98 \\ 4.5 & \text{if } SSIM > 0.98 \end{cases} \quad (2)$$

For our multi-user CLO in the mobile network, we assume that the mean utility function of the test video sequence is provided to the base station for each upstreamed video content. Figure 2 shows the utility functions for 4 different video sequences encoded at constant quality with H.264 SVC using medium-grain scalability (MGS) at CIF resolution and 30 frames/sec. More specifically, the MGS vector mode is set to zero for base layer encoding and to (3, 3, 10) for enhancement layers encoding [Schwarz et al. 2007]. The utility information is represented in the form of MOS-Rate pairs and can be signaled for instance in the RTP header extensions [Singer and Desineni 2008]. The utility can be updated for each GoP or whenever there is a large change in utility (e.g., new scene). For simplicity, we signal the utility for each video sequence at the beginning of the session. This means that the mobile terminals do not need to exchange their instantaneous utility parameters for each GoP. This reduces the signaling of media-specific information between the base station and the mobile terminals and thus reduces the overall optimization complexity.

3.2. LTE uplink model

The uplink resource allocation problem is to assign the resources (physical resource blocks (PRBs) in LTE) among the different video producers. We consider long-term optimization periods in the order of seconds. Our objective is thus to determine the average resource share (i.e., PRBs) of each user in each optimization round. This allows us to integrate our QoE-based CLO with any of the state-of-the-art schedulers for LTE uplink without the need to modify the scheduling mechanisms already deployed.

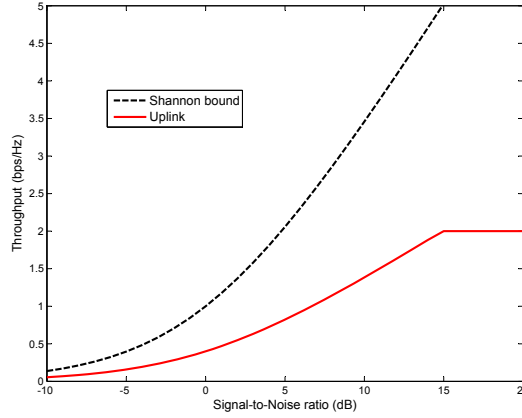


Fig. 3. Link performance level mapping of Signal-to-Noise ratio to throughput [3GPP TR 36.942 V.8.1.0 2008].

We use the link layer model from the 3GPP LTE recommendations [3GPP TR 36.942 V.8.1.0 2008] to determine the achievable throughput per PRB for a given Signal-to-Noise ratio (γ). The model from [3GPP TR 36.942 V.8.1.0 2008] approximates the throughput T in the uplink, after link adaptation and hybrid automatic repeat request, by an implementation loss $\beta = 0.4$ compared to the Shannon capacity (3) (Figure 3). As baseline uplink parameters, it further defines a γ_{min} of -10 dB, a γ_{max} of 15 dB and a maximum throughput T_{max} of 2 bps/Hz.

$$T = \begin{cases} 0 & \text{for } \gamma < \gamma_{min} \\ \beta \log_2(1 + \gamma) & \text{for } \gamma_{min} < \gamma < \gamma_{max} \\ T_{max} & \text{for } \gamma > \gamma_{max} \end{cases} \quad (3)$$

4. DISTRIBUTED QOE OPTIMIZATION

4.1. Definitions

We consider mobile terminals that are capable of generating scalable video streams which can be decomposed into several video layers (e.g., [Schwarz et al. 2007]). In addition, the mobile producers can cache parts of the video stream and upload those parts later during good channel conditions and under lower cell load. As a result, the instantaneous user uplink transmission capacity is distributed for uploading newly generated and cached video layers.

More precisely, a Group of Pictures (GoP) is encoded into a set of video layers with a common deadline (i.e., BL , EL , etc.). At each scheduling round, a new GoP (with playout deadline d_s , where d_s is the system delay for live streaming) and the layers from previous GoPs which have been cached at a mobile terminal (with playout deadline $(d_s + j \cdot t_0)$, $1 \leq j \leq n$, t_0 is the scheduling period) are considered for uplink scheduling at the mobile terminal. A schematic illustration for a scalable video stream encoded into four layers is given in Figure 4.

In addition, the following notations are used: S_L represents the set of all candidate video layers at a mobile terminal (i.e., new and cached) and $S_{L,d}$ corresponds to one set of video layers with a common deadline d . K is the number of users who have data to send. c_k is the instantaneous uplink transmission capacity, and H_k is the cache size of user k . In addition, for each video layer $l \in S_{L,d}$: L_l represents its size, R_l is the

Table I. Nomenclature

K	total number of upstream users
H_k	cache size of user k
c_k	uplink transmission capacity of user k
l, l'	video layers
$l' \prec l$	l is a descendant of layer l'
L_l	size of layer l
R_l	instantaneous rate of layer l
t_l	transmission time of layer l
ΔMOS_l	MOS improvement by transmitting layer l
d	playout deadline
$S_{L,d}$	one set of video layers with a common deadline d
S_L	all sets of video layers
a_l	binary: = 1 if layer l is scheduled
b_l	binary: = 1 if layer l can be scheduled, i.e., $a_{l'} = 1, \forall l' \prec l$

instantaneous rate, t_l is the required time to transmit the video layer, and MOS_l is its utility value which is determined according to (2). Table I summarizes the notations.

4.2. Problem formulation

The distributed QoE optimization is performed independently at each mobile terminal. The goal is to maximize the QoE by determining the set of layers to be scheduled for transmission at each round, and the rate of each scheduled layer. If layer l is scheduled, a_l is equal to 1 and 0 otherwise. $b_l \in \{0, 1\}$, describes the layer dependencies inside a GoP, i.e., if layer l is dependent on layer l' (i.e., $l' \prec l$), then b_l is equal to 1 if layer l' has been transmitted and 0 otherwise. That is, for layer l to be scheduled, layer l' must also be scheduled. ΔMOS_l denotes the additional improvement of the QoE by transmitting layer l . The distributed optimization problem for user k is formulated as:

$$\arg \max_{a_l, R_l, \forall S_{L,d}, \forall l \in S_{L,d}} \sum_{S_{L,d}} \sum_{l \in S_{L,d}} a_l \cdot b_l \cdot \Delta MOS_l(R_l) \quad (4)$$

$$s.t. \quad \sum_{S_{L,d}} \sum_{l \in S_{L,d}} a_l R_l \leq c_k, \quad (5)$$

$$\sum_{S_{L,d}} \sum_{l \in S_{L,d}} L_l(1 - a_l) \leq H_k, \quad (6)$$

$$\sum_{l \in S_{L,d}} a_l t_l \leq d, \forall S_{L,d}, \quad (7)$$

where (4) is the QoE-based objective function; (5) corresponds to the rate constraint where the sum of rates of all transmitted layers should not exceed the available capacity; (6) is a necessary condition to avoid cache overflow; the size of cached layers should not exceed the cache size; (7) is a necessary condition to transmit layers by their deadlines; the time to transmit all layers with one common deadline should be less than the deadline.

For a given set of resources c_k for user k , the problem in (4)-(7) can be solved locally for each user. In the next section, it is shown that the independent de-centralized optimization can lead to an optimal solution for each user.

4.3. Optimality analysis

The optimality analysis is composed of three steps. First, the optimization problem in (4)-(7) is transformed to a Lagrange dual problem. Then, it is shown that the dual

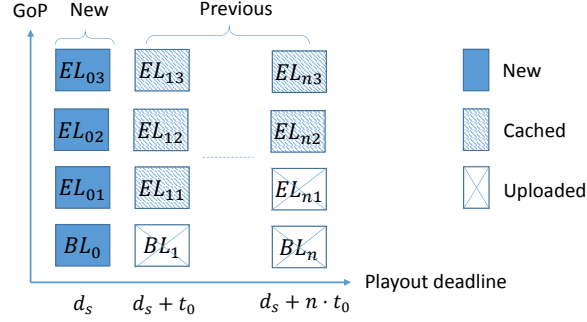


Fig. 4. A set of video layers that correspond to a new GoP and cached layers from previous scheduling rounds at a mobile terminal. The new GoP has a playout deadline d_s , which represents the system delay for live streaming. The first cached GoP has a playout deadline $d_s + t_0$, where t_0 is the scheduling interval. The n^{th} cached GoP has a playout deadline $d_s + n \cdot t_0$.

problem can be solved by an iteration method. Finally, a possible way to guarantee that the proposed solution converges to the optimal value is provided.

Step 1: Substitute R_l by $\frac{L_l}{t_l}$ in (4)-(7) and solve for the optimal values of a_l^* , t_l^* and L_l^* . From the definition of the objective function, it can be seen that (4) is a classical convex optimization problem satisfying Slater's condition [He et al. 2007]. Specifically, the Slater condition states that strong duality holds if the inequality constraints in (5)-(7) are strictly feasible, i.e., they hold with strict inequalities [Boyd and Vandenberghe 2004]. In this case, there exists an allocation policy (a_l^* , t_l^* and L_l^*), where

$$\sum_{S_{L,d}} \sum_{l \in S_{L,d}} a_l^* \frac{L_l^*}{t_l^*} < c_k, \quad (8)$$

$$\sum_{S_{L,d}} \sum_{l \in S_{L,d}} L_l^* (1 - a_l^*) < H_k, \quad (9)$$

$$\sum_{l \in S_{L,d}} a_l^* t_l^* < d, \forall S_{L,d}, \quad (10)$$

As a consequence of strong duality, the optimal duality gap is zero [Boyd and Vandenberghe 2004]. In other words, the optimal solution from the primal problem in (4)-(7) is equal to the best lower bound that can be obtained from a Lagrange dual function. As a next step, a Lagrange dual problem to achieve the optimal value is described.

Step 2: The Lagrange dual problem can be designed by augmenting the constraints from (5)-(7) into the objective function:

$$\begin{aligned} L(a_l, L_l, t_l, \phi_l, \theta_l, \delta_l) = & \sum_{S_{L,d}} \sum_{l \in S_{L,d}} a_l b_l \Delta MOS_l \left(\frac{L_l}{t_l} \right) \\ & - \sum_{S_{L,d}} \sum_{l \in S_{L,d}} \phi_l(\tau) (a_l L_l - c_k t_l) - \\ & \sum_{S_{L,d}} \sum_{l \in S_{L,d}} \theta_l(\tau) [L_l (1 - a_l) - H_k] - \sum_{l \in S_{L,d}} \delta_l(\tau) [a_l t_l - d], \end{aligned} \quad (11)$$

where, ϕ_l , θ_l , and δ_l are the Lagrange multipliers associated with the constraints of (5), (6) and (7), and τ indicates the time index ($\tau \in \mathbb{N}$).

In order to simplify the optimization problem, the Lagrangian dual problem in (11) can be further decomposed into three sub-problems [Bertsekas 1999]:

$$\max : \sum_{S_{L,d}} \sum_{l \in S_{L,d}} \phi_l(\tau) c_k t_l. \quad (12)$$

$$\max : - \sum_{S_{L,d}} \sum_{l \in S_{L,d}} \theta_l(\tau) L_l. \quad (13)$$

$$\begin{aligned} \max : & \sum_{S_{L,d}} \sum_{l \in S_{L,d}} a_l b_l \Delta MOS_l \left(\frac{L_l}{t_l} \right) - \sum_{S_{L,d}} \sum_{l \in S_{L,d}} \phi_l(\tau) a_l L_l \\ & + \sum_{S_{L,d}} \sum_{l \in S_{L,d}} \theta_l(\tau) a_l L_l - \sum_{l \in S_{L,d}} \delta_l(\tau) a_l t_l. \end{aligned} \quad (14)$$

In this case, the original problem from (11) is solved by iteratively solving the sub-problems in (12)-(14). Specifically, at each iteration, the two sub-problems in (12)-(13) can be solved independently to find the values of the variables t_l and L_l , respectively. Furthermore, sub-problem (14) can be solved to find the value of the variable a_l for a fixed t_l and L_l .

In particular, the Lagrange dual function $L_d(\phi, \theta, \delta)$ is defined as the maximum of the Lagrangian $L(a_l, L_l, t_l, \phi_l, \theta_l, \delta_l)$ over a_l , L_l and t_l for given ϕ , θ , and δ . Hence, the Lagrange dual problem can be formulated by:

$$L_d(\phi, \theta, \delta) = L \left(a_l^*(\phi, \theta, \delta), L_l^*(\phi, \theta, \delta), t_l^*(\phi, \theta, \delta) \right), \quad (15)$$

where $(\phi_l, \theta_l, \delta_l)$ are the dual variables, and $a_l^*(\phi, \theta, \delta)$, $L_l^*(\phi, \theta, \delta)$, and $t_l^*(\phi, \theta, \delta)$ correspond to the values a_l , t_l and L_l when achieving the optimal solution. As explained in Step 1, since the duality gap is equal to zero, the Lagrange dual function returns the optimal solution for the optimization problem in (4)-(7).

Step 3: Since L_d may be non-differentiable, an iterative sub-gradient method can be used to solve it by introducing the variables ρ_ϕ , ρ_θ , and ρ_δ . Please note that L_d is a convex optimization problem since the objective to be maximized is concave and can be solved by a conventional KKT method [Chiang 2005]. For each dual variable, the sub-gradient method generates a sequence of feasible points which are updated at each iteration τ . Specifically, from (12):

$$\phi_l(\tau + 1) = [\phi_l(\tau) + \rho_\phi(\tau) g_\phi(\tau)]^+ = \left[\phi_l(\tau) + \rho_\phi(\tau) \sum_{S_{L,d}} \sum_{l \in S_{L,d}} c_k t_l \right]^+, \quad (16)$$

where $g_\phi(\tau)$ is the sub-gradient of (12) at $\phi_l(\tau)$, and $\rho_\phi(\tau)$ represents the iteration step size.

Similarly, from (13):

$$\theta_l(\tau + 1) = [\theta_l(\tau) + \rho_\theta(\tau) g_\theta(\tau)]^+ = \left[\theta_l(\tau) - \rho_\theta(\tau) \sum_{S_{L,d}} L_l \right]^+. \quad (17)$$

where $g_\theta(\tau)$ is the sub-gradient of (13) at $\theta_l(\tau)$, and $\rho_\theta(\tau)$ represents the iteration step size.

ALGORITHM 1: Low-complexity distributed algorithm for scalable video transmission at mobile terminals

input : available transmission capacity c_k for user k
output: transmission vector $\underline{a} \forall S_{L,d}, \forall l \in S_{L,d}$
Initialize transmission vector $\underline{a} = \{0\}$;
Drop layers with expired delay;
while $c_k > 0$ **do**
 if this is the first iteration **then**
 for all candidate GoPs **do**
 if c_k is enough to send the layer l in GoP j that has not been sent yet and with the highest priority **then**
 calculate utility improvement $U_j = \Delta MOS_l(R_l)/(R_l)$;
 else
 $U_j = 0$;
 end
 end
 else
 only update $U_n = \Delta MOS_l(R_l)/(R_l)$, l is the layer in GoP n with the second highest priority;
 end
 find $\arg \max_n U_j$ where n is the GoP with the maximum utility improvement; set $a_l = 1$;
 $c_k = c_k - R_l$;
end
Check cache overflow, drop layers if necessary;
Output transmission vector $\underline{a}$;

In addition, $\frac{\partial(14)}{\partial a_l} = 0$ at the optimal solution,

$$\begin{aligned} & \sum_{S_{L,d}} \sum_{l \in S_{L,d}} b_l \Delta MOS_l \left(\frac{L_l}{t_l} \right) - \sum_{S_{L,d}} \sum_{l \in S_{L,d}} \phi_l(\tau) L_l + \\ & \sum_{S_{L,d}} \sum_{l \in S_{L,d}} \theta_l(\tau) L_l - \sum_{l \in S_{L,d}} \delta_l(\tau) t_l = 0. \end{aligned} \quad (18)$$

Hence, $\delta_l(\tau)$ can be determined as a function of the iteration step size $\rho_\delta(\tau)$:

$$\begin{aligned} & \delta_l(\tau + 1) = \\ & \left[\delta_l(\tau) + \rho_\delta(\tau) \sum_{S_{L,d}} \sum_{l \in S_{L,d}} \left(b_l \Delta MOS_l \left(\frac{L_l}{t_l} \right) \right) \right]^+ \end{aligned} \quad (19)$$

$\rho_\phi(\tau)$, $\rho_\theta(\tau)$ and $\rho_\delta(\tau)$ represent the iteration step size. Certain choices of step sizes, such as $\rho_\phi(\tau) = \frac{\rho_1}{\tau}$, $\rho_\theta(\tau) = \frac{\rho_2}{\tau}$, and $\rho_\delta(\tau) = \frac{\rho_3}{\tau}$, where $\rho_1, \rho_2, \rho_3 > 0$ guarantee that this algorithm will converge to the result obtained via joint optimization. An exemplary illustration of the choices of the iteration step sizes for determining the upper bound of the convergence step is provided in Appendix A..

4.4. Low-complexity distributed algorithm

The distributed QoE optimization problem from (4)-(7) can be solved using the interior point method [Boyd and Vandenberghe 2004]. Nevertheless, the time complexity for determining the optimal solution will increase as a function of the number of cached video layers. As a result, searching through all the possible transmission policies of

the newly generated and cached GoPs could become infeasible for real-time scheduling. In this section, a low-complexity iterative greedy algorithm is proposed to solve the problem in (4)-(7) at each mobile terminal. The optimization problem can be further simplified by considering that: 1) the GoPs are independent and thus the layer scheduling decisions in one GoP do not affect the subsequent GoPs. 2) The dependencies between the video layers inside a GoP can be described by an acyclic graph [Chou and Miao 2006]. In our case, the layers are represented by a sequential acyclic graph (BL , EL , etc.). An iterative greedy approach is considered for allocating the resources among the different GoPs. The objective is to determine which video layers should be scheduled/cached at each optimization round.

A description of the proposed low-complexity distributed algorithm is given in Algorithm 1. At a given scheduling round, there are S_L video layers whose transmission policies are defined by $a_l \in \{0, 1\}, \forall S_{L,d}, \forall l \in S_{L,d}$. The transmission vector $\underline{a}$ is initialized to $\underline{0}$. At each iteration, the GoP which provides the maximum utility improvement is allocated a fraction of resources necessary to transmit a new layer l . Inside a GoP, the transmission priorities are determined by the acyclic graph. As a result, the iteration complexity is determined by the number of candidate GoPs. The above procedure is repeated until no more layers can be transmitted with the rate constraint. At the end of the optimization, the layer with the minimum MOS improvement is dropped in the case of cache overflow.

5. MULTI-USER CROSS-LAYER OPTIMIZATION

To determine the uplink resources provided to each mobile terminal (i.e., c_k in (5)), we use the long-term abstraction model proposed in [Saul et al. 2007]. The model defines the actual data rate c_k for user k as a function of its resource share α_k and its maximum achievable rate $C_{max,k}$ if all the PRBs are allocated exclusively to user k .

$$c_k = f_k(\alpha_k) = \alpha_k C_{max,k} \quad 0 \leq \alpha_k \leq 1, \forall k \quad (20)$$

The objective of the multi-user CLO is to determine the resource share α_k of each user that maximizes the overall QoE. The objective function, that maximizes the sum of utilities of K users, can then be described by:

$$\arg \max_{(\alpha_1, \dots, \alpha_K)} \sum_{k=1}^K U_k(c_k) \quad \text{subject to} \quad \sum_{k=1}^K \alpha_k = 1 \quad (21)$$

Each α_k value corresponds to the fraction of total PRBs assigned to user k . U_k represents the average utility function of user k (Figure 2). At each optimization round, a new $C_{max,k}$ is estimated using (3) based on the average channel quality indicator (CQI) of user k . A greedy resource allocation algorithm, similar to the work in [Jurca and Frossard 2007], is used to determine the value of α_k . It is initialized by assigning an equal amount of resources to every user. The algorithm iteratively takes a small amount of resources from the user who is less sensitive to the decrease in resources and assigns it to the user who gets the maximum benefit, until no further improvement in (21) is possible. The greedy algorithm has a low computational complexity and achieves indeed a result which is very close to a full search strategy [Thakolsri et al. 2009].

To realize our QoE optimization framework, the following signaling information is considered. The playout deadlines (Live and VoD) can be communicated before the start of the streaming session. Mobile terminals inform the base station about their utility function U_k as explained in Section 3.1. The available capacity for each terminal c_k is provided by the base station to the mobile terminal. This is updated in each

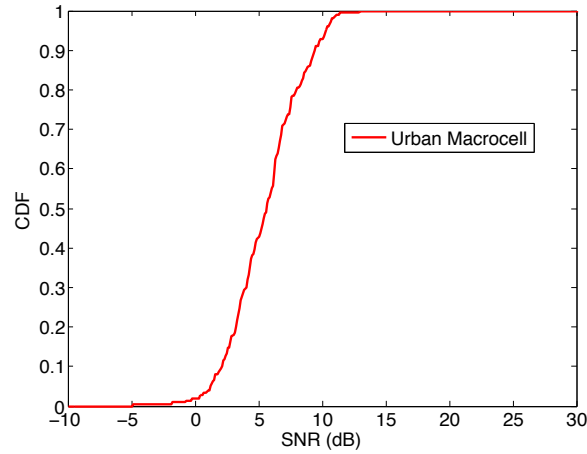


Fig. 5. Distribution of uplink SNR in an urban macrocell [3GPP TR 36.814 V.9.0.0 2010]. The distributions are used for the calibration of simulations so that different simulators meet standard evaluation references. The figure is reproduced using the LTE simulator from [Ellenbeck 2012].

optimization round. This means that the base station periodically (i.e., each 266 msec) informs the mobile terminal about its target transmission capacity. No further signaling or resource allocation is required at the lower layers. By constraining the resource budget of each terminal to the available resources in the network, our approach proactively avoids the system running into overload.

6. RESULTS

6.1. Intention of study

The performance evaluation is realized in two steps. First, a Matlab-based LTE simulator is considered to validate and study the computational complexity of our proposed QoE optimization framework. Second, simulation results using an LTE OPNET simulator are conducted. Specifically, we 1) exploit the advantages of our CLO compared to the case where the resources are only determined by the standard LTE PF scheduler, and 2) analyze the potential gains of applying the distributed QoE optimization at a mobile terminal by observing the perceived video quality across the live and on-demand consumers.

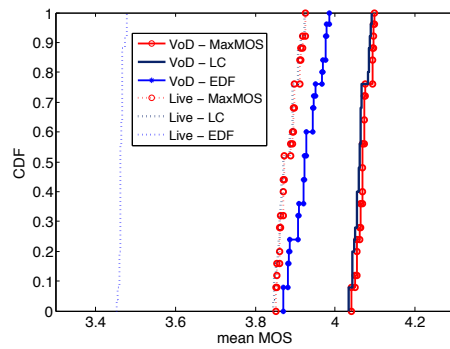
6.2. Performance and complexity analysis

A Matlab-based simulator for LTE uplink with a similar simulation setup to [El Essaili et al. 2011b] is used. We consider three mobile users upstreaming different video content over the LTE network. The channel realizations of the video producers are tuned to follow the typical SNR distribution of the users for uplink in an urban macrocell [3GPP TR 36.814 V.9.0.0 2010] (Figure 5). These distributions represent the average SNR a user will experience in the uplink. The uploaded videos are requested by a live consumer (delay 266 msec) and a video-on-demand (VoD) consumer (delay 10 sec). The simulation parameters are summarized in Table II. Moreover, the following schemes are compared:

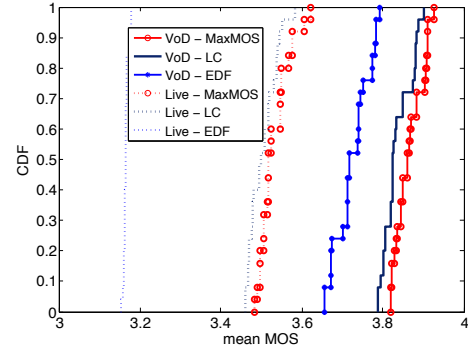
- **MaxMOS**: This represents the optimal solution for the distributed optimization problem in (4)-(7) which is determined using the interior point method [Boyd and

Table II. Simulation setup

Parameter	Value
LTE bandwidth	5 MHz
Number of PRBs	25
Number of subcarriers	300
PRB size	12 subcarriers
Link layer model [3GPP TR 36.942 V.8.1.0 2008]	see (3)
Channel model	Urban macrocell [3GPP TR 36.814 V.9.0.0 2010]
CQI averaging cycle	266 msec
Number of upstream users	3
Video sequences	<i>Football, Bus and Soccer</i>
Relative cache size	0.3
Live delay	1 GoP (266 msec)
VoD delay	10 sec
Video codec	H.264/SVC, CIF, 30 fps, 4 layers
Simulation time	10 sec
Simulation runs	25
Simulator	LTE Matlab simulator [?]



(a) SSIM-MOS mapping



(b) PSNR-MOS mapping

Fig. 6. Mean MOS of all users that correspond to the optimal solution from (4)-(7), the approximate solution from the proposed low complexity scheme, and the earliest deadline first approach. The performance of the low-complexity solution is close to the optimal one and superior to the EDF approach for both live and on-demand consumption. Both PSNR and SSIM based QoE mapping models are considered.

Table III. Average gains on the MOS scale

	PSNR-MOS model	SSIM-MOS model
Live (EDF $\rightarrow$ QoE)	0.36	0.4
VoD (EDF $\rightarrow$ QoE)	0.11	0.14
QoE (Live $\rightarrow$ VoD)	0.33	0.2
EDF (Live $\rightarrow$ VoD)	0.56	0.46

Vandenberghe 2004]. A multi-user cross-layer optimization is applied that maximizes the overall utility according to (21).

- **LC**: This represents the low complexity solution for the distributed optimization problem as described in Section 4.4. A similar multi-user cross-layer optimization as in MaxMOS scheme is applied.
- **Earliest deadline first (EDF)**: The base station serves the user with the earliest deadline. Each video producer performs a distributed video layer optimization with the objective of transmitting the layers with the most imminent deadline first.

Table IV. Mean number of iterations per optimization round for the optimal approach and the proposed low complexity scheme. The number of iterations to achieve the solution can be drastically reduced as the number of cached video layers increases.

Number of layers	Number of iterations MaxMOS	Number of iterations LC
5	6	3.2
10	14.5	3.9
15	48.8	2.8
20	64.9	3.35
25	109.2	3.73
30	148.6	3.64
35	293.2	3.73
40	586.9	3.36

Figure 6 further depicts the cumulative distribution function (CDF) of the mean MOS of all users. The MOS for the LC scheme is close to the optimal solution for both live and VoD transmission. The slight MOS reduction depends on the choice of enhancements layers that are uploaded for on-demand consumption. Furthermore, the QoE schemes improve the video quality compared to the EDF scheme as they select the video layers that provide the highest MOS improvement.

In addition, Figure 6 shows the CDF of the mean MOS when a PSNR-based mapping model is considered. Specifically, the PSNR is linearly mapped to the MOS, where MOS can take on any value between 1.0 (25 dB) and 4.5 (40 dB), which represent the worst and best QoE, respectively. The results from the PSNR-based mapping conform with the SSIM-based ones: First, both figures show similar MOS gains for the QoE-aware approaches compared to the EDF scheduling scheme, for both live and on-demand consumption cases. In addition, noticeable MOS improvements are noted for each scheme for the on-demand versus live transmission. Also, the performance of the low complexity approach is very close to the optimal MaxMOS scheme for both mapping models. The average gains are further summarized in Table III.

Table IV compares the computational complexity in terms of number of iterations per optimization round for the low-complexity approach and the MaxMOS scheme. It is clear that the computational complexity can be significantly reduced with the proposed low complexity approach. In fact, it suffices to search for the GoP with the highest utility gain and then use the acyclic graph to select the corresponding layer inside the GoP to be scheduled. As a result, the number of iterations per optimization round is proportional to the number of scheduled layers for each user. Moreover, the mean execution time per optimization round is $127msec$ for the MaxMOS scheme and $0.78msec$ for the low-complexity approach.

6.3. Simulation setup

For the following evaluations, we use an LTE OPNET simulator. Different from the previous section which focuses on long-term distributed and centralized optimizations at the mobile terminals and in the mobile network, the OPNET simulator includes a short-term PF scheduler which allows us to run both optimizations on top of the underlying LTE scheduler. Indeed, the Matlab-based simulations represent a special and ideal case where the result of the CLO exactly matches the rate allocation by the scheduler.

We consider a resource-constrained LTE cell with 4 users upstreaming the *Bus*, *Soccer*, *Football* and *Foreman* test video sequences. The videos are encoded at constant quality with the H.264 scalable video codec [Schwarz et al. 2007] into 4 layers using MGS scalability and have a CIF resolution and a frame rate of 30 frames per second. Please note that for each new GoP, the average rate and distortion characteristics of the different layers within this GoP are considered. Furthermore, we consider a rela-

Table V. Simulation parameters

Application parameters	
Video codec	H.264 SVC [Schwarz et al. 2007]
SVC scalability	MGS, 4 layers at CIF resolution and 30 fps
GoP size	8 frames
Video sequences	<i>Football, Bus, Soccer</i> and <i>Foreman</i>
Relative cache size	0.3
Live delay	1 GoP (266 msec)
VoD delay	20 sec
LTE parameters	
Transmission scheme	SC-FDMA
Base frequency	1920 MHz
Bandwidth	5 MHz
Bandwidth per PRB	180 KHz
Subcarrier spacing	15 KHz
eNodeB antenna gain	15 dBi
UE speed	30 km/h
UE antenna gain	0 dBi
Max Tx power per user	24 dBm
UE receiver sensitivity	-160 dBm
Scheduler	Proportional fair
Channel model	Urban macrocell
Shadowing	disabled
CQI averaging cycle	266 msec
Simulation parameters	
Number of users	4
Simulation runs	50
Simulation time	100 sec
Simulator	LTE OPNET 16.0

tive cache size of 30% at the mobile terminals. In other words, a mobile terminal can cache 30% of its video without dropping any layer. Without loss of generality, we measure the MOS of the uploaded videos at the media server for live consumers (delay of 1 GoP) and for on-demand consumers (delay of 20 sec). The simulation parameters are summarized in Table V.

Moreover, we compare between our cross-layer optimized and the standard non-optimized proportional fair scheduling. In both schemes, a distributed QoE optimization is applied at each mobile terminal for joint live and on-demand video layer scheduling, given the available network capacity. The low-complexity distributed approach from Section 4.4 is considered herein. We refer to *LC w CLO* and *LC w/o CLO* to represent both schemes, respectively.

6.4. Simulation results

We first consider a single simulation run with the 4 video producers undergoing dynamic channel variations as depicted in Figure 7. We analyze the temporal video quality of all users for both *LC w CLO* and *LC w/o CLO* schemes.

The PF scheduler (Figure 8(a)-8(d)) assigns the resources by considering the channel characteristics and irrespective of the uploaded video content. This results in high MOS for the less demanding videos while more demanding ones or users moving to the edge of the cell will suffer in streaming performance. For instance, the *Bus* and *Soccer* producers undergo bad channel situations between 160-190 sec and 130-150 sec, respectively. This leads to a severe degradation in the perceived video quality of the uploaded videos. Also, for the *Football* producer, the MOS will only improve after 140 sec as the user moves towards the center of the cell. Meanwhile, the less demanding *Foreman* producer who is in less favorable channel conditions can maintain a satisfactory video quality throughout the streaming session.

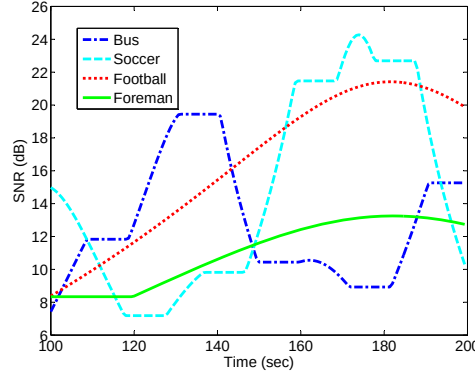


Fig. 7. Channel variations of the 4 video producers as a function of simulation time. The channel traces are obtained from the LTE OPNET simulator with users moving at a speed of 30 km/h in an urban macrocell.

The *LC w CLO* scheme (Figure 8(e)-8(h)) distributes the resources by considering the channel and the video characteristics of each producer. This means that users undergoing temporary periods of bad channel conditions (e.g., *Bus* and *Soccer*) can benefit from the QoE optimization to upload a better video quality during channel fades. Furthermore, the demanding *Football* producer can achieve a higher MOS compared to the PF non-optimized case. This improvement comes at the expense of a slight degradation in the MOS performance for the *Foreman* producer.

Figure 8 further illustrates the gains of the distributed QoE optimization at the mobile terminals for both schemes. In both cases, a refined streaming performance is observed for all on-demand video consumers. Interestingly, the live quality of the *soccer* video in Figure 8(f) drops as the upstream user moves towards the edge (115 sec). In this case, only the base layer is uploaded for live consumption. As the user moves again to the center of the cell (135 sec), the live video quality improves again and additionally the enhancement layers of previous GoPs are uploaded for on-demand consumption. For a VoD consumer, who requests the video from the media server after 20 seconds, the MOS will be substantially ameliorated (MOS increases by up to 1.2).

Figure 9 depicts the mean MOS of all the uploaded videos for the same simulation run. The *LC w CLO* scheme sustains the average video quality and distributes the resources among the video producers to maximize the overall quality of experience. Meanwhile, the *LC w/o CLO* scheme only adapts to the channel conditions which results in a fluctuating video quality. Moreover, both schemes show the potential improvement of applying the distributed QoE optimization at the mobile terminals. By caching and uploading the enhancement layers, the overall user experience for video on-demand consumption is enhanced.

Figure 10 shows the average mean MOS and the distribution of the mean MOS over 50 simulation runs. As can be seen, for both schemes, the mean MOS for on-demand viewers can be improved by uploading additional enhancement layers before the play-out deadline. In addition, the *LC w CLO* scheme manages to improve the mean MOS substantially for both live and on-demand consumers, when compared to the *LC w/o CLO* scheme. In fact, by adapting the transmission rates of the different producers, our CLO prevents the system from running into overload and thus allows for an improved overall user satisfaction. Besides, it results in less variation in the resulting mean MOS across the different simulation runs. Again, by shaping the transmission capacity of each terminal a more consistent user experience can be achieved.

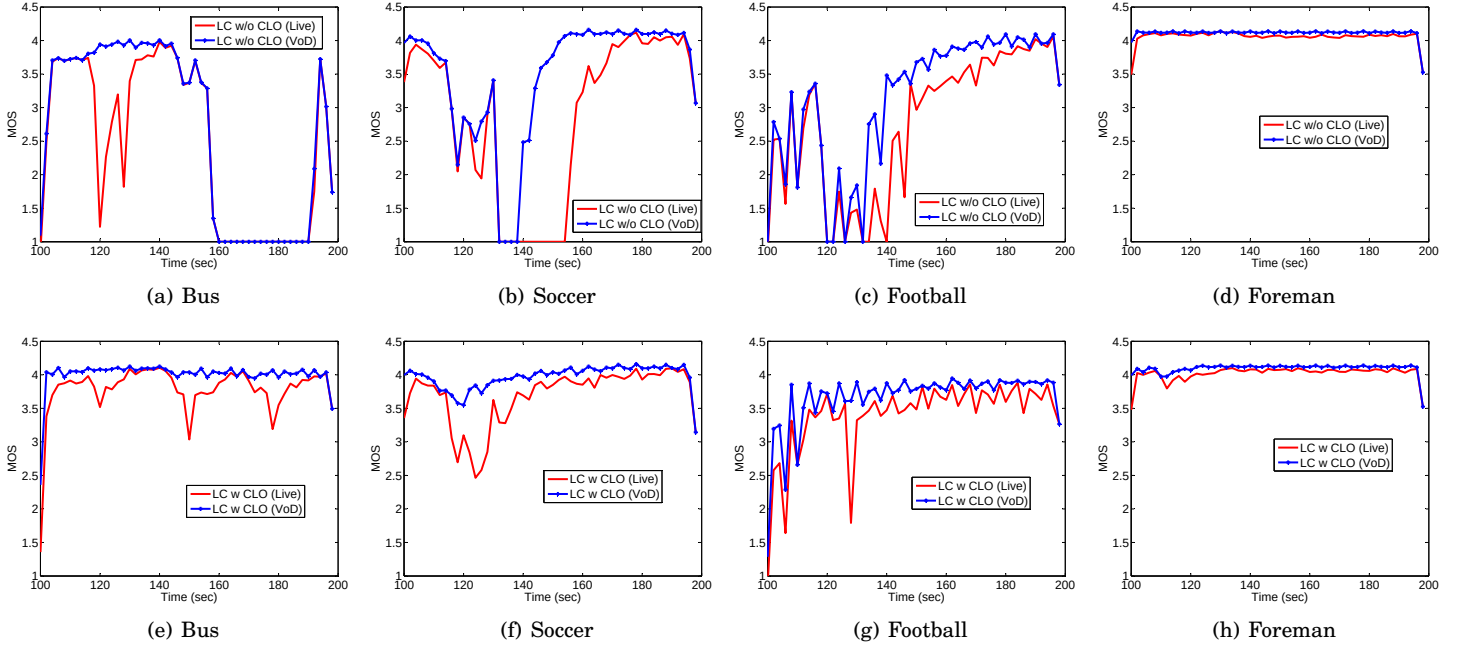


Fig. 8. MOS of the different users as a function of simulation time: (a)-(d) applying distributed QoE optimization at the mobile producers and proportional fair scheduling in the network; (e)-(h) applying both distributed and CLO on top of the PF scheduler.

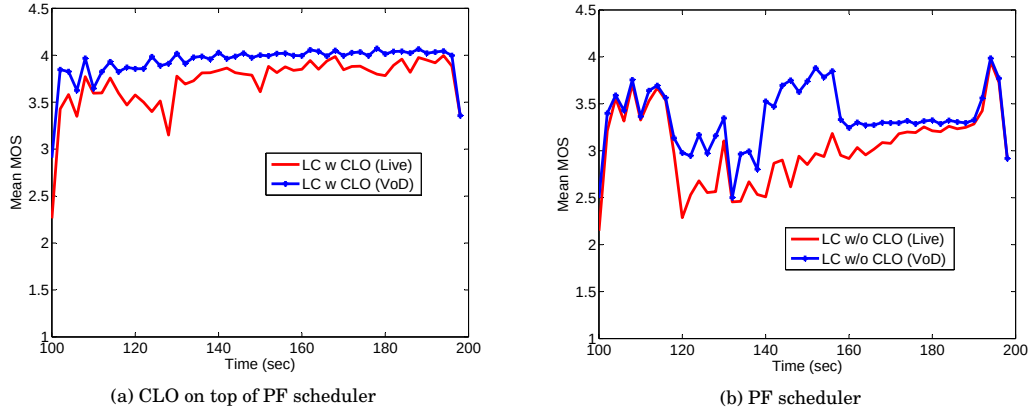


Fig. 9. Mean MOS of all users as a function of simulation time. Our proposed CLO provides a better and smoother video quality compared to the standard LTE scheduler.

6.5. Discussion

In this paper, we particularly address the radio access network (RAN) where the limited wireless resources are shared among multiple users. The majority of streaming users use non-GBR bearers and RAN user plane congestion is mainly caused by video transmission [3GPP TR 23.705 2013]. Our proposed solution is particularly interesting for cellular operators when the system is overloaded. First, instead of blocking the users suffering from bad radio conditions, our solution allows admitting those users

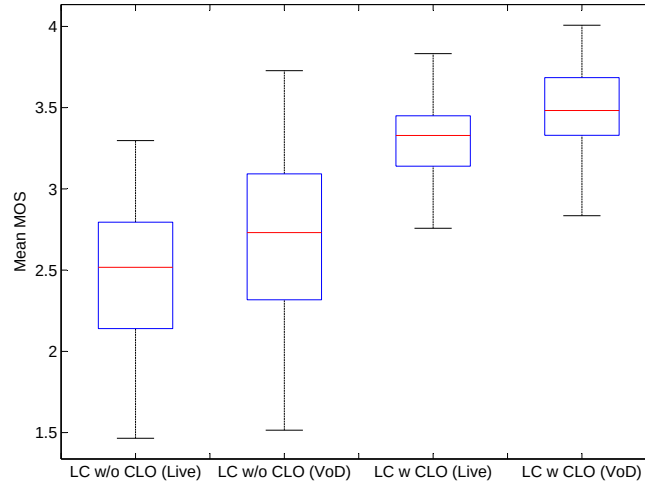


Fig. 10. Average mean MOS of all users over 50 simulation runs. The error bars represent the worst and best runs. The boxplot shows the median, 25th and the 75th percentiles.

while achieving graceful QoE degradation for the less resource-demanding users. Second, for edge users who are already admitted to the system our approach avoids critical quality degradations in video quality. The approach is applied for LTE as it represents the state-of-the-art mobile technology and also considers OFDM time-slotted resource allocation similar to WiMAX. Nevertheless, we believe that it is easy for other researchers to extend the proposed approaches to other mobile systems.

One of the distinguishing factors of our solution is that the optimization is carried out in the MOS domain and is independent of the QoE mapping model. This would ideally allow us to distinguish across contents in terms of their perceived video quality. The presented results encourage the development of better QoE models [Dosselmann and Yang 2011] by showing that the gains that can be achieved by QoE-driven optimizations will be high. The presented solutions do not account for the temporal quality fluctuations across different GoPs. This can be, however, easily integrated into the optimization function as proposed in [Thakolsri et al. 2011]. With the constant quality setting in SVC, we show that there is indeed very little quality variations from one GoP to another.

The cross-layer optimization and the distributed QoE optimization presented in this paper can be implemented in real-time and require minimal signaling between the base station and the mobile equipments. Scalable video coding results in higher throughput overhead compared to non-scalable video coding, only if one representation is needed. For live and on-demand streaming, we need at least two representations in order to serve live users with a basic quality and be able to enhance the video quality for on-demand users. The overhead of independent non-scalable coding of multiple representations is higher than that of SVC. A main challenge for the practical application of the proposed approach is the availability of real-time SVC encoders in the mobile devices. This should be realistic in the near future given the rapid improvements in the processing and storage capabilities of mobile phones. Recently, the call for device-centric architectures for 5G networks has surfaced [Boccardi et al. 2014]. The objective is to enrich mobile devices with smart caching, transmission and decision-making capabilities. Our distributed optimization approach is one step in this direction.

7. CONCLUSIONS

This paper studies the uplink video transmission from mobile terminals over fourth generation mobile networks. Specifically, we propose a QoE-based uplink resource optimization system which involves both the network and the mobile producers. We present an optimal distributed model and a low-complexity iterative algorithm for scalable video transmission at mobile terminals, which adapts to the variable channel characteristics to provide optimal user experience for different types of consumers. We validate both theoretically and practically our proposed QoE optimization approach for an LTE uplink system. Significant gains in terms of resource efficiency and user experience are illustrated.

The conducted evaluations show that the content agnostic LTE scheduler results in suboptimal performance, in particular for resource-demanding videos. By acquiring general knowledge about the content characteristics, we proactively shape the wireless resources to improve the level of user satisfaction in the mobile cell. Moreover, this work exploits an important use-case for scalable video transmission over wireless networks. By properly adapting the media transmission to the available network resources, significant improvements in perceived video quality can be obtained over time for the upstreamed videos.

A. CONVERGENCE BOUND

The relationship between the convergence steps and ρ_1, ρ_2, ρ_3 is studied by analyzing their dependencies when reaching the optimal solution. Specifically,

1) When $\frac{\partial(12)}{\partial t_l} = 0$:

$$\sum_{S_{L,d}} \sum_{l \in S_{L,d}} \phi_l(\tau) c_k = 0. \quad (22)$$

2) When $\frac{\partial(13)}{\partial L_l} = 0$:

$$\sum_{S_{L,d}} \sum_{l \in S_{L,d}} \theta_l(\tau) = 0. \quad (23)$$

3) When $\frac{\partial(14)}{\partial t_l} = 0$:

$$\sum_{S_{L,d}} \sum_{l \in S_{L,d}} -a_l b_l \Delta MOS \left(\frac{L_l}{t_l} \right) L_l \frac{1}{t_l^2} - \sum_{l \in S_{L,d}} \delta_l(\tau) a_l = 0. \quad (24)$$

More precisely, from (22) and (23), $\phi_l(\tau) = 0$ and $\theta_l(\tau) = 0$ at the optimal solution, respectively. Combining with (16) and (17), the convergence bound is independent of the parameters ρ_1 and ρ_2 .

Furthermore, by replacing $\rho_\delta(\tau) = \frac{\rho_3}{\tau}$ and including $\delta_l(\tau)$ from (19) into (24):

$$\begin{aligned} & \sum_{S_{L,d}} \sum_{l \in S_{L,d}} -a_l b_l \Delta MOS \left(\frac{L_l}{t_l} \right) \frac{L_l}{t_l^2} - \\ & \sum_{l \in S_{L,d}} \left[\delta_l(\tau - 1) + \frac{\rho_3}{\tau} \sum_{S_{L,d}} \sum_{l \in S_{L,d}} \left(b_l \Delta MOS \left(\frac{L_l}{t_l} \right) - c_k t_l + L_l - t_l \right) \right]^+ a_l = 0. \end{aligned} \quad (25)$$

Moreover, when optimality is achieved, $\delta_l(\tau) = \delta_l(\tau - 1) = C$, where C is a constant. So,

$$\sum_{S_{L,d}} \sum_{l \in S_{L,d}} -a_l b_l \Delta MOS \left(\frac{L_l}{t_l} \right) \frac{L_l}{t_l^2} - \rho_3 \sum_{S_{L,d}} \sum_{l \in S_{L,d}} (b_l \Delta MOS \left(\frac{L_l}{t_l} \right) - c_k t_l + L_l - t_l) = C \quad (26)$$

$$\Rightarrow \sum_{S_{L,d}} \sum_{l \in S_{L,d}} -a_l b_l \Delta MOS \left(\frac{L_l}{t_l} \right) \frac{L_l}{t_l^2} - C = \rho_3 \sum_{S_{L,d}} \sum_{l \in S_{L,d}} (b_l \Delta MOS \left(\frac{L_l}{t_l} \right) - c_k t_l + L_l - t_l) \quad (27)$$

$$\Rightarrow \rho_3 \propto \frac{\sum_{S_{L,d}} \sum_{l \in S_{L,d}} -a_l b_l \Delta MOS \left(\frac{L_l}{t_l} \right) \frac{L_l}{t_l^2}}{\sum_{S_{L,d}} \sum_{l \in S_{L,d}} (b_l \Delta MOS \left(\frac{L_l}{t_l} \right) - c_k t_l + L_l - t_l)} \quad (28)$$

Furthermore, the MOS improvement is bounded and $b_l \Delta MOS \left(\frac{L_l}{t_l} \right) \leq 4.5$. Therefore, the upper bound of the convergence step is $O \left(\frac{\sum_{S_{L,d}} \sum_{l \in S_{L,d}} a_l \frac{L_l}{t_l^2}}{\sum_{S_{L,d}} \sum_{l \in S_{L,d}} (c_k t_l - L_l)} \right)$.

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